SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSA16 Data Warehousing and Data Mining

Assignment 2 – CO2

CO2 Coverage: *Data mining from a database perspective, data mining functionalities, steps in the data mining process (KDD), architecture and classification of data mining systems, basic data mining tasks and issues, data dispersion, data cleaning, integration, transformation, reduction and discretization.*

Course Outcome: CO2 – Analyse data dispersion techniques and list the various algorithms available for mining the data.

Nature: Individual | **Duration:** 1 Week | **Total Marks:** 50

Q1 – Matching Business Problems to Mining Functionalities — 5 Marks

Problem Context – Telecom Subscriber Analytics

TeleNova is a telecom operator with 8 million subscribers. Its analytics team has received the following six business problems:

- **P1:** Find which plans and add-on services customers typically purchase together.
- **P2:** Estimate whether a given customer will leave the network next month.
- **P3:** Divide subscribers into natural usage segments, without any predefined labels.
- **P4:** Identify SIM cards whose calling behaviour is very different from all others.
- **P5:** Summarise the general characteristics of high-value customers and show how they differ from low-value customers.
- **P6:** Study how monthly data usage of customers has changed over the last 24 months.

Your Challenge

For each problem, identify:

- The **data mining functionality** involved (concept/class description, association, classification, prediction, clustering, outlier analysis or evolution analysis)
- Whether it is **supervised or unsupervised**, and the kind of pattern or knowledge produced
- The data attributes required and the data source (database or warehouse)
- At least two **candidate algorithms** from the syllabus and the reason for choosing them

Design Decision

The marketing manager proposes buying a single “all-purpose mining algorithm” for all six problems. Evaluate this proposal using your analysis.

Evidence Expected

A table of six problems mapped to functionality, learning type, data and algorithms, and a reasoned evaluation of the single-algorithm proposal.

Q2 – The KDD Process, System Architecture and Mining Issues — 5 Marks

Problem Context – Hospital Readmission Mining

MediCore, a hospital network, wants to discover factors that cause patients to be readmitted within 30 days. Patient data is spread across admission, laboratory, pharmacy and billing systems.

Your Challenge

- Map each KDD step (selection, preprocessing, transformation, mining, interpretation/evaluation) to **specific activities** MediCore must perform
- Draw the architecture of a typical data mining system for MediCore (database/warehouse server, knowledge base, mining engine, pattern-evaluation module, user interface) and mark where each requirement is handled
- Classify the system on four views: kinds of data, kinds of knowledge, techniques used and applications
- Identify **at least five data mining issues** specific to this scenario (methodology, user interaction, efficiency and scalability, diversity of data types, privacy/social) and a mitigation for each

Inquiry

A hospital director says: “We already have SQL and statistics, so we do not need data mining.” Using an example from MediCore, explain how a mining task differs from a database query and from a basic statistical summary.

Evidence Expected

KDD mapping, architecture diagram, system classification, issue–mitigation table and a reasoned answer to the director’s statement.

Q3 – Describing and Comparing Data Dispersion **7 Marks**

Problem Context – E-Commerce Delivery Times

An e-commerce company recorded the delivery time (in minutes) of 24 orders from one dark store:

28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95

Your Challenge

- Compute the **mean, median, mode, midrange**, and comment on what the relationship between mean and median suggests about the distribution
- Compute the **range, quartiles Q1 and Q3, IQR, variance and standard deviation**, showing your method (state which quartile convention you use)
- Prepare the **five-number summary** and draw a **boxplot**; apply the **1.5 × IQR rule** to identify outliers
- Remove the largest value and recompute all measures. Which measures changed the most, and why?
- Classify mean, median, variance, standard deviation and range as **distributive, algebraic or holistic** measures, and explain what this means when the data is split across many warehouse partitions
- Verify your manual results using a tool (spreadsheet, Python, Weka or KNIME statistics node) and explain any difference

Inquiry

The operations team wants to promise customers a “typical” delivery time. Which measure would you recommend and why? What would you report in addition, and why?

Evidence Expected

Manual calculations, boxplot, outlier identification, measure classification, tool verification and a reasoned recommendation.

Q4 – Cleaning Noisy and Incomplete Data **6 Marks**

Problem Context – Smart-Meter Energy Readings

A utility collects hourly readings from household smart meters. A sample extract is shown below:

Meter	Household type	kWh	Voltage (V)	Region
M101	Apartment	5	230	North

Meter	Household type	kWh	Voltage (V)	Region
M102	Apartment	(missing)	231	North
M103	House	38	229	north
M104	House	-14	232	South
M105	(missing)	19	2300	South
M106	Apartment	11	230	S.

The sorted kWh values of 12 other hourly readings are: **5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38**

Your Challenge

- Identify **every defect** in the extract and classify it (missing value, noise, inconsistency, invalid value)
- For each defect, propose and justify a handling strategy from: ignore the tuple, fill manually, global constant, attribute mean, class-wise mean, or most probable value (regression/inference)
- Apply **equal-frequency binning (depth 4)** to the 12 sorted values and smooth by **bin means** and by **bin boundaries**, showing all bins
- Explain how regression and clustering could be used to detect or smooth noise, and why a reading such as 2300 V must be treated differently from a reading of 5 kWh

Inquiry

Replacing every missing value with the global mean is quick. Explain **how this affects the dispersion of the attribute and the quality of patterns mined later**, and when it should be avoided.

Evidence Expected

Defect classification, strategy table, binning calculations and a reasoned discussion on the impact of imputation on dispersion and mining.

Q5 – Integration and Transformation of Data **7 Marks**

Problem Context – University Data Consolidation

A university combines its Admissions database (attributes: *student_id, name, dob, hostel_status, scholarship*) with its Learning Management System (attributes: *sid, full_name, attendance_pct, marks, scholarship_flag*).

Your Challenge

Part A – Integration

- Identify the entity-identification, naming, unit/format and value conflicts and show how they are resolved in the integrated schema
- Explain why integration often creates **redundant attributes**, and how correlation analysis helps detect them

Part B – Redundancy analysis

- For attendance % = {60, 70, 75, 80, 85, 90} and marks = {52, 58, 66, 70, 79, 85}, compute the **Pearson correlation coefficient** and interpret it
- For the observed counts below, compute the **χ^2 statistic** (with expected values) and state whether *hostel status* and *scholarship* are independent at the 0.05 level (critical value for 1 degree of freedom = 3.84)

	Scholarship	No scholarship	Total
Hostel	120	180	300
Day scholar	80	320	400
Total	200	500	700

Part C – Transformation

- Annual family income (₹) of five students: **20,000; 35,000; 50,000; 80,000; 150,000**. Normalise using **min-max (0 to 1), z-score and decimal scaling**
- Two students have (age, income) = (19, 35,000) and (21, 80,000). Compute their Euclidean distance **before and after** min-max normalisation of both attributes (assume age range 17–25 and income range 20,000–150,000) and explain what this reveals for distance-based mining
- Give one example each of **aggregation, generalisation and attribute construction** for this data

Evidence Expected
Conflict-resolution table, correlation and χ^2 calculations with interpretation, three normalisation methods and a before/after distance comparison.

Q6 – Data Reduction and Discretization — 7 Marks

Problem Context – Industrial Sensor Monitoring

A factory logs **200 sensor attributes** from machines, producing **5 million rows** per month. Analysts need faster mining without losing the ability to detect machine failure.

Your Challenge

Part A – Choosing reduction strategies

- For each category, describe **one technique and when it is suitable** for this data: data-cube aggregation, attribute subset selection (forward selection, backward elimination, decision-tree induction), dimensionality reduction (PCA), numerosity reduction (sampling, histograms, clustering) and compression
- Compare **simple random sampling without replacement, with replacement, cluster sampling and stratified sampling** and state which is appropriate if only 0.5% of the rows are failures
- Explain how many principal components you would keep and the criterion you would use

Part B – Discretization by calculation

Temperature readings: 3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44

- Discretize into 4 intervals using **equal-width** and **equal-frequency** binning. Show the intervals and the number of values in each
- Using the labelled values below, find the best **entropy-based split point** (candidate split points are midpoints between adjacent values; show the information gain of each candidate):

Value	10	12	15	18	22	25	28	31	35	40
Class	N	N	N	N	Y	Y	N	Y	Y	Y

- Compare the unsupervised and supervised discretization results and explain which suits a classification task.

Evidence Expected
Reduction strategy justification, sampling comparison, equal-width/equal-frequency intervals, information-gain table and a reasoned comparison.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate — 13 Marks

Problem Context – Loan-Default Pipeline Failure

A retail bank built a model to predict loan default. Its preprocessing pipeline was:

1. Missing *income* values were replaced by the global mean income.
2. The attribute *occupation* (18% missing) was dropped.
3. *Income* (highly right-skewed) was discretized into 5 **equal-width** intervals.

4. All numeric attributes were min-max normalised using the **complete dataset**, before splitting into training and test sets.
5. A random 10% sample of rows was used for training. Only 3% of loans in the data are defaults.

The model reported **97% accuracy** on the test set. After deployment, it missed most defaulters and the bank lost money.

A. Diagnose the Computational Bottleneck – 2 Marks

Identify the defects in the pipeline. For each, name the **data preprocessing principle violated** and explain how it can mislead the model or the evaluation.

B. Define the Design Objective – 2 Marks

State measurable objectives (for example the metric to improve and the acceptable trade-off) and the constraints that the redesigned pipeline must satisfy.

C. Design the Revised Preprocessing Strategy – 4 Marks

Design an **ordered** pipeline that addresses each defect using appropriate cleaning, integration, transformation, reduction and discretization techniques. For each step, justify the technique chosen over at least one alternative and state whether it is computed on training data only.

D. Predict Before Implementing – 1 Mark

Before running anything, predict how your redesign will change: the dispersion of *income*, the class distribution in the training data, and accuracy versus recall on defaulters.

E. Implement and Validate the Redesign – 2 Marks

Implement the original and the redesigned pipelines on a suitable dataset (for example UCI *German Credit* or a similar public credit dataset) using Weka, KNIME or Python. Compare them with the same test data and report suitable metrics.

F. Defend the Final Design – 2 Marks

Compare your predictions with the results. Defend your final pipeline, and state one risk that remains.

Evidence Expected

A diagnosis, measurable objectives, a justified pipeline, a written prediction, validated comparison results and a defence of the final design.

MARK DISTRIBUTION

Question	Assessed Area	Marks
Q1	Mining Functionalities and Algorithm Mapping	**5**
Q2	KDD Process, Architecture and Issues	**5**
Q3	Data Dispersion	**7**
Q4	Data Cleaning	**6**
Q5	Data Integration and Transformation	**7**
Q6	Data Reduction and Discretization	**7**
Q7	**Day-4 Design Challenge – Diagnose, Redesign and Validate**	**13**
TOTAL		**50**

Submission Requirements

Submit:

- Your assignment report containing the required evidence for Q1–Q7
- Spreadsheets/code/workflows used for calculations and validation
- Screenshots of tool outputs (Weka/KNIME/Python) with the settings used
- Datasets (or links) used in Q7

Manual calculations are **primary evidence**. Tool output may be used to verify your work but must not replace the working shown.

Assessment Verification

Students may be asked to explain any part of their:

- Manual calculations and their method
- Choice of preprocessing techniques
- Predictions and results
- Design Challenge

during a brief verification discussion/viva. The student should therefore be able to **explain and defend the evidence submitted as their own work**.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook definitions or tool outputs.

It assesses your ability to:

Understand → Describe → Clean → Transform → Reduce → Predict → Validate → Defend

Your conclusions and design decisions must be supported by **your own reasoning and evidence**.

Q1 – MATCHING BUSINESS PROBLEMS TO MINING FUNCTIONALITIES (5 MARKS)

TeleNova has 8 million subscribers. Its six problems require different kinds of knowledge. No single mining algorithm is appropriate for every task because association discovery, labelled prediction, segmentation, anomaly detection, summarisation and temporal evolution have different objectives and data requirements.

Problem	Functionality / learning type	Knowledge produced	Required attributes and source	Candidate algorithms and reason
P1: Plans and add-ons bought together	Association analysis; unsupervised	Frequent itemsets and rules such as {Plan A} → {Data Booster}; support, confidence and lift.	Subscriber/order-level basket, plan ID, add-on IDs, purchase date; billing/CRM warehouse.	Apriori: clear frequent-itemset generation and understandable support/confidence. FP-Growth: avoids repeated candidate generation and is more scalable for large transaction sets.
P2: Predict customer leaving next month	Classification / prediction; supervised	Churn probability or churn/not-churn label for each subscriber.	Historical churn outcome, tenure, recharge frequency, complaints, dropped calls, data usage trend, plan changes, payment failures; customer warehouse with time-correct labels.	Decision Tree: interpretable rules for retention teams. Random Forest: combines trees and often improves robustness. Logistic Regression is a useful probability baseline.
P3: Natural usage segments without labels	Clustering; unsupervised	Groups such as high-data/low-voice, low-usage, and frequent-roaming segments.	Monthly voice minutes, data GB, SMS, recharge amount, roaming, tenure, add-ons; subscriber usage warehouse.	K-means: efficient for large numeric datasets after scaling. DBSCAN: can identify dense groups and leave unusual users as noise, but needs parameter tuning.
P4: SIMs with unusual calling behaviour	Outlier analysis; usually unsupervised	Anomaly score or ranked SIMs with unusual call volume, destinations, duration or timing.	Calls per hour/day, unique numbers, international calls, call duration distribution, night-call ratio, location changes; CDR database/warehouse.	Isolation Forest: efficient anomaly scoring in high-volume data. Local Outlier Factor (LOF): detects users unusual relative to local peers.
P5: Characteristics of high-value vs low-value customers	Concept/class description; can be descriptive or supervised after labels are defined	Profiles and distinguishing attribute ranges, e.g. high-value customers have high tenure and regular recharge.	ARPU, lifetime value, tenure, recharge, complaint rate, usage, plan type and a defined value segment; CRM/warehouse.	Decision-tree rule extraction: gives readable distinguishing conditions. Attribute-oriented induction / OLAP profiling: summarizes characteristics by class. Association rules can reveal common attribute combinations.
P6: Monthly data usage evolution over 24 months	Evolution analysis / time-series trend analysis; generally descriptive, may use supervised forecasting if target future usage is labelled	Trend, seasonality, change points and forecast of monthly usage.	Subscriber ID, month, data GB, plan changes, campaign dates, network events; time-series warehouse.	Moving averages / trend analysis: reveal direction and smoothing. ARIMA: models temporal dependence and forecasts when

				assumptions and series diagnostics are suitable.

1.1 Supervised and unsupervised distinction

Supervised learning uses examples with a known target label. P2 needs historical churn labels; P5 can become supervised if high-value/low-value is explicitly defined as a target class. Unsupervised learning does not require target labels: P1 discovers co-occurrence, P3 discovers groups, and P4 ranks anomalies without a complete labelled anomaly set. P6 primarily studies temporal behaviour; forecasting can be supervised in the sense that past observations are used to predict a future target, but trend/evolution description does not require class labels.

1.2 Evaluation of the all-purpose algorithm proposal

The proposal should be rejected. Apriori produces association rules, not a calibrated churn probability. A decision tree requires a target label and does not automatically discover natural clusters. K-means assigns cluster membership but does not explain which SIMs are isolated anomalies. Time-series methods model temporal order, which is lost if monthly usage is treated as an unordered feature vector. TeleNova should select an algorithm per business objective, establish common data preparation and model governance, and compare candidate algorithms using task-appropriate metrics.

- Association: support, confidence and lift.
- Churn classification: recall, precision, F1, ROC-AUC or PR-AUC; emphasize recall and cost-weighted evaluation.
- Clustering: silhouette score, stability and business interpretability.
- Outliers: precision at a review capacity, analyst validation and false-positive rate.
- Evolution: forecast error such as MAE/RMSE and trend stability.

A common mining platform may host several algorithms, but that is not the same as one universal algorithm.

Q2 – THE KDD PROCESS, SYSTEM ARCHITECTURE AND MINING ISSUES (5 MARKS)

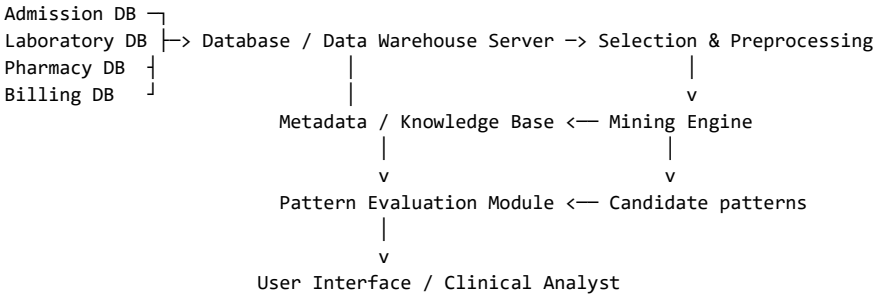
MediCore wants to discover factors associated with a patient being readmitted within 30 days. Admission, laboratory, pharmacy and billing records must be integrated. The objective is to identify useful patterns for clinical review, not to automatically claim that a pattern proves causation.

2.1 KDD steps mapped to MediCore

KDD step	MediCore activities	Output / control
1. Selection	Define cohort: eligible discharged patients; choose observation period and 30-day follow-up window. Select admission/discharge records, diagnosis, age band, lab results, medication, length of stay, prior admissions and billing/utilization features.	Cohort table with a stable patient and encounter key. Exclude records without sufficient follow-up or document censoring.
2. Preprocessing	Resolve duplicate patient IDs; correct invalid dates; standardize diagnosis and medication codes; handle missing lab values; investigate impossible values; remove duplicate claims and leakage fields recorded after the prediction time.	Clean integrated records plus a quality/rejection log.
3. Transformation	Construct target readmitted_within_30_days; derive prior_admission_count, length_of_stay, abnormal_lab_count, medication_count and time-since-previous-discharge; encode categorical values and scale numeric fields if needed.	Mining-ready feature table with clear prediction time and target definition.
4. Data mining	Train classification models such as decision tree, logistic regression or random forest; optionally cluster patient profiles for exploratory analysis.	Rules, feature importance, risk scores and patient groups.

5. Interpretation / evaluation	Evaluate on a time-separated test set; inspect recall, precision, PR-AUC and calibration; review patterns with clinicians; check subgroup performance and clinical plausibility.	Validated findings, limitations and recommendations for human review.

2.2 Typical data mining system architecture



Database/warehouse server stores integrated historical data. The knowledge base contains medical terminology, domain constraints, concept hierarchies and thresholds. The mining engine executes classification, association, clustering or other algorithms. The pattern-evaluation module ranks patterns by predictive usefulness, novelty and validity. The user interface lets analysts select cohorts, parameters, inspect patterns and export reports. Privacy, role-based access and audit logging apply across all layers.

2.3 Classification of the MediCore mining system

View	Classification for MediCore
Kinds of data	Relational clinical and administrative databases; integrated warehouse tables; time-stamped admission, lab, pharmacy and billing records; structured categorical and numeric data.
Kinds of knowledge	Classification rules and risk scores, associations among conditions/treatments/utilization, clusters of patient profiles, temporal patterns and outlier cases.
Techniques used	Database and statistical analysis, machine learning, classification, clustering, association analysis, visualization and domain knowledge.
Applications	Hospital quality improvement, discharge planning, readmission risk review, resource planning and clinical audit.

2.4 Data mining issues and mitigations

Issue category	Scenario-specific risk	Mitigation
Methodology / data quality	Different systems use inconsistent patient IDs, coding standards and event dates.	Master patient index, terminology mapping, validation rules, documented cohort and target definitions.
User interaction	Clinicians may not understand model rules or may distrust black-box scores.	Clinician co-design, interpretable rules, explanation of limitations and feedback workflow.
Efficiency and scalability	Large lab and billing histories create expensive joins and feature computation.	Warehouse indexing/partitioning, incremental feature tables, sampling for exploration and scalable batch processing.
Diversity of data types	Numeric labs, diagnosis codes, medication records and time series have different structures.	Type-specific cleaning and encoding; preserve temporal order; use appropriate feature extraction.
Privacy / social	Sensitive health data could expose patients or create unfair treatment across groups.	Minimum necessary access, de-identification where appropriate, encryption, audit logs, consent/legal review and subgroup fairness checks.
Data leakage / validity	A feature entered after discharge may reveal the outcome or information unavailable at prediction time.	Set a prediction cut-off and include only data available before that time; use temporal train/test separation.

2.5 Why SQL and basic statistics are not enough

A SQL query can count readmissions by diagnosis, and a statistical summary can report the average length of stay. These are useful but usually answer predefined questions. Data mining searches a large feature space for patterns that are not explicitly specified in advance and evaluates whether they generalize to unseen patients. For example, a decision-tree model may discover an interaction: patients with a prior admission, a particular combination of abnormal laboratory indicators and a long stay have elevated 30-day readmission risk. The model can test this pattern on a later time period and estimate its predictive performance. SQL can retrieve the data and statistics can describe it; mining combines feature discovery/model learning and predictive validation. The discovered association is not automatically causal and must be reviewed clinically.

Q3 – DESCRIBING AND COMPARING DATA DISPERSION (7 MARKS)

Data (minutes), already sorted: 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95. There are n = 24 observations. Quartile convention: median-of-halves (Tukey hinges). Variance is shown as both population and sample variance; use population values if these 24 readings are the complete group of interest, and sample values if they represent a sample.

3.1 Central tendency

Sum = 1143 minutes. Mean = 1143 / 24 = 47.625 minutes. Median = average of the 12th and 13th values = (43 + 45) / 2 = 44 minutes. Modes = 35 and 42 minutes (each occurs twice). Midrange = (minimum + maximum) / 2 = (28 + 95) / 2 = 61.5 minutes.

Mean > median (47.625 > 44), suggesting a positively/right-skewed distribution. The high value 95 pulls the mean upward more than it affects the median.

3.2 Dispersion and quartiles

Lower half (first 12): 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43. Q1 = average of its 6th and 7th values = (36 + 38)/2 = 37. Upper half (last 12): 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95. Q3 = (53 + 55)/2 = 54. IQR = 54 − 37 = 17. Range = 95 − 28 = 67.

Population variance $\sigma^2 = \Sigma(x - \text{mean})^2 / n = 213.3177$ min²; population standard deviation $\sigma = \sqrt{213.3177} = 14.6054$ min. Sample variance $s^2 = \Sigma(x - \text{mean})^2 / (n-1) = 222.5924$ min²; sample standard deviation s = 14.9195 min.

3.3 Five-number summary and outlier test

Minimum	Q1	Median	Q3	Maximum
28	37	44	54	95

IQR fences: lower = Q1 − 1.5(IQR) = 37 − 25.5 = 11.5; upper = Q3 + 1.5(IQR) = 54 + 25.5 = 79.5. Any observation outside [11.5,79.5] is a potential outlier. Therefore 95 minutes is an upper outlier; no lower outliers.

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Boxplot (schematic): 28 ── [ Q1=37 | Median=44 | Q3=54 ] ── 72        •95
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The upper whisker ends at 72, the largest observation within the upper fence. The value 95 is plotted separately as an outlier point.

3.4 Recalculation after removing 95

Measure	All 24 values	After removing 95	Change
Mean	47.625	45.5652	−2.0598
Median	44	43	−1
Mode	35 and 42	35 and 42	No change
Midrange	61.5	50	−11.5
Range	67	44	−23
Q1	37	36	−1
Q3	54	52.5	−1.5
IQR	17	16.5	−0.5
Population variance	213.3177	120.7675	−92.5502
Population SD	14.6054	10.9894	−3.6160
Sample variance	222.5924	126.2569	−96.3355
Sample SD	14.9195	11.2364	−3.6831

The range and midrange change sharply because both depend directly on the maximum. Variance and standard deviation also fall substantially because they give weight to distance from the mean. The median and quartiles are more resistant to an extreme value; the mode is unchanged.

3.5 Classification of measures for distributed warehouse partitions

Measure	Class	Reason
Mean	Algebraic	Can be combined from partial count and partial sum: global mean = total sum / total count.
Median	Holistic	Exact global median generally cannot be derived from a fixed-size set of partition medians; values/order information or an approximate quantile sketch is needed.
Variance	Algebraic	Can combine count, sum and sum of squares (or stable partial count/mean/M2 statistics).
Standard deviation	Algebraic	Derived by combining variance state and taking square root.
Range	Distributive	Global minimum = min(partition minima), global maximum = max(partition maxima); global range = max − min.

Distributive measures can be aggregated directly from partition summaries. Algebraic measures can be computed from a fixed number of summary values. Holistic measures may require information that grows with the data size or approximate algorithms.

3.6 Tool verification and recommendation

Verification can be performed in Excel or Python. In Python, use mean(), median(), mode(), min()/max(), quantile methods and var(ddof=0 or 1). Quartile conventions differ: NumPy's default percentile interpolation may not equal the median-of-halves convention used manually, so select a matching method or explain the difference. Population versus sample variance also changes the denominator.

For a customer-facing 'typical' delivery promise, recommend the median (44 minutes) because it is resistant to the unusually long 95-minute delivery and better represents the middle order. Also report the 75th percentile (Q3 = 54 minutes) or an on-time percentage within a stated target, because customers and operations need to understand slower deliveries. A median alone does not describe the service tail.

Q4 – CLEANING NOISY AND INCOMPLETE DATA (6 MARKS)

The supplied extract contains six meter rows and a separate sorted set of 12 kWh readings. Data cleaning must distinguish a missing or noisy value from an impossible value; automatic replacement can conceal faults.

4.1 Defects in the sample extract

Meter	Defect	Classification	Recommended handling
M102	kWh missing	Missing value	Impute only if required for analysis; use class-wise/household-type mean or a regression estimate based on comparable time, household and meter history. Keep an imputation flag.
M103	Region is 'north' while standard value is 'North'	Inconsistency	Normalize case using a controlled region lookup.
M104	kWh = −14 for ordinary consumption	Invalid value / possible noise	Check meter reset, export/net-metering semantics and source fault. Quarantine until validated; do not replace blindly. If negative values are

			impossible by business rule, mark invalid.
M105	Household type missing	Missing value	Infer from trusted customer/meter master if available; otherwise use Unknown category or exclude from analyses requiring household type.
M105	Voltage = 2300 V versus expected around 230 V	Likely noisy / invalid value	Check decimal/scale/unit and sensor fault. Confirm source. If it is a misplaced decimal, correction must be supported by source evidence; otherwise quarantine.
M106	Region = 'S.'	Inconsistent abbreviation	Map to 'South' only if a controlled region code table confirms it; otherwise flag for review.

The brief does not provide a valid range or engineering tolerance for voltage. Therefore 2300 V is a strong anomaly relative to the other rows, but it should be verified against the meter specification rather than corrected by assumption. A value of 5 kWh can be a valid low-consumption hour; it is not inherently invalid.

4.2 Equal-frequency binning (depth 4) and smoothing

Sorted values: 5, 9, 11, 14, 18, 19, 23, 26, 27, 30, 33, 38. Depth 4 means four values per bin, giving 12/4 = 3 bins if interpreted as four bins of equal frequency? The assignment wording says 'depth 4'; in standard equal-frequency binning, depth is the number of values per bin. Therefore use 3 bins of 4 values each. If the instructor intended 4 bins, the alternative is 4 bins of 3 values; both are shown to remove ambiguity.

Interpretation	Bin 1	Bin 2	Bin 3	Bin 4
4 bins of 3 values (often expected for 4 bins)	5, 9, 11	14, 18, 19	23, 26, 27	30, 33, 38
Depth 4 = 4 values per bin	5, 9, 11, 14	18, 19, 23, 26	27, 30, 33, 38	—

For the four-bin interpretation (3 values per bin), bin means are: (5+9+11)/3 = 8.33; (14+18+19)/3 = 17; (23+26+27)/3 = 25.33; (30+33+38)/3 = 33.67. Smoothing by bin means replaces each value in its bin by the corresponding mean.

Bin	Original values	Smoothed by mean	Boundaries	Smoothed by boundaries
1	5, 9, 11	8.33, 8.33, 8.33	5 and 11	5, 11, 11
2	14, 18, 19	17, 17, 17	14 and 19	14, 19, 19
3	23, 26, 27	25.33, 25.33, 25.33	23 and 27	23, 27, 27
4	30, 33, 38	33.67, 33.67, 33.67	30 and 38	30, 30, 38

If depth 4 is followed literally (three bins of four), bin means are 9.75, 21.5 and 32.0. Boundary smoothing gives [5,14] → 5,5,14,14; [18,26] → 18,18,26,26; [27,38] → 27,27,38,38. State which interpretation is used in the final submission and confirm with the instructor.

4.3 Regression and clustering for noise

Regression can estimate an expected reading from related variables such as the same meter's previous and next hours, household type, temperature or day-of-week. A large residual between observed and predicted values can flag noise. Regression should not be used to overwrite a measurement until the source is checked; preserve original and imputed values separately.

Clustering groups similar meter profiles. A reading or meter profile far from its cluster centroid or density region may be flagged as anomalous. Clustering can help detect unusual patterns across many readings, but an anomaly is not automatically an error: a household may genuinely have high usage. Domain rules and validation are required.

4.4 Why global-mean imputation can be harmful

Replacing missing values with one global mean reduces the attribute's variance because every missing record is placed exactly at the centre. It can weaken correlations, hide subgroup differences and create an artificial pile-up around the mean. Later clustering may make groups look more similar than they really are, while classification may learn distorted relationships. Avoid global-mean imputation when data are missing not at random, when groups have very different distributions, when missingness itself carries information, or when values are safety-critical. Prefer domain-based inference, class-wise estimates, regression/multiple imputation or an explicit missing category/indicator, and retain an imputation flag.

Q5 – INTEGRATION AND TRANSFORMATION OF DATA (7 MARKS)

5.1 Part A – Integration conflicts

Conflict type	Admissions source	LMS source	Resolution in integrated schema
Entity identification	student_id	sid	Map both to a governed student_key using an authoritative crosswalk; validate one-to-one or investigate duplicates.
Naming conflict	name	full_name	Standardize to student_name; retain source lineage.
Format conflict	dob may be date text in one format	Potentially different date encoding	Parse to ISO date YYYY-MM-DD using source locale and validate plausible dates.
Value conflict	hostel_status	No direct equivalent	Standardize to controlled values: Hostel / Day Scholar / Unknown.
Scholarship conflict	scholarship	scholarship_flag	Map to a common scholarship_status after checking whether both represent the same meaning and time period.
Unit / scale conflict	No direct equivalent	attendance_pct and marks	Store attendance as percentage 0–100 and marks with documented maximum/scale; do not compare marks until scales are aligned.

Integration can create redundant attributes when two sources describe the same entity or measure under different names. For example, scholarship and scholarship_flag may encode the same fact. Correlation analysis can identify highly related numeric attributes, while dependency checks and domain review determine whether one is redundant. Correlation alone does not prove that two attributes are interchangeable.

5.2 Part B – Pearson correlation

Attendance = {60,70,75,80,85,90}; marks = {52,58,66,70,79,85}. Pearson $r = \Sigma[(x_i-\bar{x})(y_i-\bar{y})] / \sqrt{[\Sigma(x_i-\bar{x})^2\ \Sigma(y_i-\bar{y})^2]}$. Using the six paired observations gives $r = 0.9851$ (approximately). This is a very strong positive linear association: higher attendance is associated with higher marks in this small sample. It does not prove that attendance alone causes higher marks; sample size is only six and other factors may affect marks.

5.3 Chi-square test of independence

Observed table: Hostel: scholarship 120, no scholarship 180, row total 300. Day scholar: scholarship 80, no scholarship 320, row total 400. Column totals are 200 and 500; grand total N = 700.

Cell	Observed O	Expected E = row total × column total / 700
Hostel, scholarship	120	300×200/700 = 85.714
Hostel, no scholarship	180	300×500/700 = 214.286
Day scholar, scholarship	80	400×200/700 = 114.286
Day scholar, no scholarship	320	400×500/700 = 285.714

$\chi^2 = \Sigma (O-E)^2/E = 33.60$. Degrees of freedom = (2−1)(2−1) = 1. Critical value at 0.05 = 3.84. Since 33.60 > 3.84, reject the null hypothesis of independence. In this sample, hostel status and scholarship status are statistically associated. This does not establish causation.

5.4 Part C – Normalisation of annual family income

Income values: ₹20,000; ₹35,000; ₹50,000; ₹80,000; ₹150,000. Minimum = 20,000; maximum = 150,000; range = 130,000. Min-max formula: x' = (x−min)/(max−min).

Income ₹	Min-max 0–1	Z-score (population σ)	Decimal scaling (j=6)
20,000	0.0000	−0.995	0.02
35,000	0.1154	−0.735	0.035
50,000	0.2308	−0.475	0.05
80,000	0.4615	0.045	0.08
150,000	1.0000	2.160	0.15

For z-score, mean = ₹67,000 and population standard deviation ≈ ₹50,251.87; z = (x−67,000)/50,251.87. Decimal scaling divides by 10^j, where j is the smallest integer such that max(|x'|) < 1. Since maximum income is 150,000, j = 6.

5.5 Euclidean distance before and after min-max scaling

Students A=(age 19, income ₹35,000) and B=(age 21, income ₹80,000). Before scaling: $d = \sqrt{[(21-19)^2 + (80{,}000-35{,}000)^2]} = \sqrt{(4 + 2{,}025{,}000{,}000)} \approx 45{,}000.0000$. Income dominates because its numeric scale is much larger than age.

Using age range 17–25 and income range ₹20,000–₹150,000: A'=(0.25, 0.1154), B'=(0.50, 0.4615). Distance after scaling = $\sqrt{[(0.50-0.25)^2 + (0.4615-0.1154)^2]} \approx 0.42699$. After normalisation, age and income contribute on comparable 0–1 scales. This is important for distance-based methods such as K-means and nearest neighbours.

5.6 Aggregation, generalisation and attribute construction

- Aggregation: calculate average attendance per department or total scholarship amount per academic year.
- Generalisation: replace individual date of birth with age band (18–20, 21–23, 24+) or city with district/state.
- Attribute construction: calculate attendance_band, age_at_admission or scholarship_to_income_ratio from existing attributes.

Q6 – DATA REDUCTION AND DISCRETIZATION (7 MARKS)

6.1 Part A – Reduction strategies for 200 attributes and 5 million rows/month

Reduction category	Technique and application	Suitability / caution
Data-cube aggregation	Aggregate sensor readings by machine, hour, shift or day; retain min, max, mean, standard deviation and failure count.	Useful for trend dashboards and routine mining; preserve high-frequency windows around failures because averages can hide spikes.
Attribute subset selection	Forward selection adds useful features; backward elimination removes weak features; decision-tree induction selects split attributes.	Reduces irrelevant/redundant sensors. Use training data only and validate feature selection inside cross-validation to avoid leakage.
Dimensionality reduction	PCA transforms correlated sensor attributes into principal components.	Useful when sensors are highly correlated; components may be less interpretable and scaling is essential.
Numerosity reduction	Sampling, histograms, clustering or representative prototypes.	Reduces rows while retaining distribution patterns; rare failures need special protection.

Compression	Lossless compression for exact raw retention; controlled lossy compression for approximate summaries where permitted.	Lossless is safest for audit/failure diagnosis. Lossy compression must preserve peaks and fault signatures.

6.2 Sampling methods and rare failures

Sampling method	How it works	Strength / weakness
Simple random without replacement	Each row has equal chance; selected row cannot be selected again.	No duplicates; can miss rare failures when prevalence is only 0.5%.
Simple random with replacement	Each draw is independent; a row may appear multiple times.	Easy statistical properties but duplicates reduce unique information.
Cluster sampling	Select whole groups, e.g. machines, sites or time blocks.	Convenient when data are naturally grouped; may miss failure diversity if selected clusters are not representative.
Stratified sampling	Split into strata such as failure/non-failure and sample from each.	Ensures rare failure cases are represented; appropriate here, with weights used when estimating population-level metrics.

With only 0.5% failures, a random sample may contain too few failure examples for reliable model training. Stratified sampling is preferred for a training/evaluation dataset because it preserves or deliberately increases representation of the minority class. For final evaluation, keep a test set that reflects the real 0.5% prevalence, or use appropriate weights and report the actual prevalence.

6.3 How many principal components to retain

Do not choose an arbitrary fixed number. Standardize numeric sensor attributes, fit PCA on training data, and retain the smallest number of components that explains a preselected cumulative variance target (for example 90–95%), then validate failure recall/PR-AUC and reconstruction error. For safety-critical failure detection, retain additional components if they improve rare-failure detection even when the variance threshold is already reached. High-variance directions are not guaranteed to contain the most predictive failure signal.

6.4 Part B – Equal-width discretization

Temperature values: 3, 7, 7, 9, 12, 14, 14, 15, 18, 21, 21, 22, 25, 25, 27, 31, 36, 36, 40, 44. Minimum = 3; maximum = 44; range = 41. Four equal-width intervals have width = 41/4 = 10.25.

Interval	Values	Count
[3, 13.25)	3, 7, 7, 9, 12	5
[13.25, 23.5)	14, 14, 15, 18, 21, 21, 22	7
[23.5, 33.75)	25, 25, 27, 31	4
[33.75, 44]	36, 36, 40, 44	4

Equal-width bins have equal numeric span but unequal counts (5, 7, 4, 4), which is expected when the distribution is not uniform.

6.5 Equal-frequency discretization

Bin	Values	Count
1	3, 7, 7, 9, 12	5
2	14, 14, 15, 18, 21	5
3	21, 22, 25, 25, 27	5
4	31, 36, 36, 40, 44	5

Equal-frequency bins contain the same number of observations, but their numeric widths differ. The repeated 21 is split across the second and third bins under this simple rank-based partition; a tie-preserving implementation could keep equal values together, producing slightly unequal counts.

6.6 Entropy-based supervised split

Class labels in order: N, N, N, N, Y, Y, N, Y, Y, Y. There are five N and five Y, so initial entropy $H(S)=-0.5\log_2 0.5-0.5\log_2 0.5=1$ bit. For each candidate midpoint t, split values ≤t and >t. Information Gain = $H(S) - (|L|/10)H(L) - (|R|/10)H(R)$.

Candidate midpoint	Information gain (bits)
11.0	0.1080
13.5	0.2365
16.5	0.3958
20.0	0.6100
23.5	0.2781
26.5	0.1245
29.5	0.3958
33.0	0.2365
37.5	0.1080

The best split is at 20.0, with information gain ≈ 0.6100 bits. It separates the first four N observations from the remaining six observations, which contain four Y and two N. It is the best single binary split among the candidate midpoints, but further recursive splitting may be needed to classify all records.

6.7 Unsupervised versus supervised discretization

Equal-width and equal-frequency binning are unsupervised because they use only the distribution of the attribute and ignore class labels. Entropy/information-gain discretization is supervised because it chooses boundaries that improve class separation. For a classification task, supervised discretization can be more useful because the intervals are selected with respect to the target class. It can overfit if too many splits are selected, so pruning and cross-validation are required.

Q7 – DIAGNOSE, REDESIGN AND VALIDATE (13 MARKS)

The bank's pipeline achieves 97% test accuracy but misses most defaulters. Since only 3% of loans are defaults, a classifier that predicts every loan as non-default already obtains 97% accuracy. Therefore accuracy alone is misleading. The redesign must focus on minority-class detection and a leakage-free evaluation.

7A – Diagnose the computational bottleneck / pipeline defects (2 marks)

Original step	Defect / violated principle	How it misleads
Global mean income imputation	Ignores skew, groups and missingness mechanism; can reduce variance and distort relationships.	Creates artificial central values, hides income distribution and may weaken default-risk signals.
Drop occupation with 18% missing	Drops potentially predictive information without testing missingness or usefulness.	Occupation may relate to income stability and default; missingness itself may be informative.
Five equal-width income bins	Poor choice for highly right-skewed income; intervals may be imbalanced and compress important low-income detail.	Most records may fall in a small number of bins while high-income ranges are sparse; boundary effects lose information.
Normalize before train/test split	Data leakage: test-set min/max influence training transformation.	Evaluation indirectly uses information from test data and may be optimistic; preprocessing must be fitted on training data only.
Random 10% training sample with 3% defaults	Severe class imbalance and too few positive cases; sampling not stratified.	Training sample may contain very few defaults, so the model learns majority class and has poor default recall.
97% accuracy used as success criterion	Metric mismatch for rare positive class and asymmetric financial costs.	An all-non-default model can score 97% while detecting zero defaulters.

7B – Measurable design objectives and constraints (2 marks)

- Primary objective: improve default-class recall at a controlled precision/false-positive rate, evaluated on an untouched test set with real class prevalence.
- Report PR-AUC, recall, precision, F1, confusion matrix, ROC-AUC, balanced accuracy and calibration; do not use accuracy alone.

- Choose a decision threshold using the bank's cost of missed defaults versus cost of manual review, not automatically 0.5.
- Constraints: no target leakage; transformations fitted only on training folds; preserve customer privacy; maintain reproducibility; explain predictions; avoid unjustified removal of potentially predictive features.
- Use stratified splitting for development and a time-based holdout if loan origination dates are available, so future data are not used to predict the past.

7C – Revised ordered preprocessing pipeline (4 marks)

Order	Revised step	Justification and alternatives	Fit on training only?
1	Define target and prediction timestamp; remove post-outcome fields.	Prevents target leakage; only information available at application/decision time is allowed.	Definition fixed before split; feature construction respects timestamp.
2	Split into train/validation/test using stratification; prefer chronological holdout when appropriate.	Keeps test data untouched; random split is less realistic when deployment predicts future loans.	Split first.
3	Inspect missingness and occupation; add missing indicator and impute occupation with most-probable/class-aware model or 'Unknown' category.	Retains occupation information instead of dropping 18%; compare with dropping only as a baseline.	Imputer/encoder fitted on training data only.
4	Impute income using median or a robust model, optionally within meaningful groups; add income_missing indicator; inspect outliers and log-transform skewed income.	Median is robust to right skew; global mean is sensitive to high incomes. Regression imputation may preserve relationships but can overfit.	Fit imputer and log/transform parameters on training only.
5	Handle numeric outliers using domain validation and robust transformation; do not automatically delete legitimate high incomes.	Winsorization or robust scaling can limit extreme influence; deleting valid high earners loses information.	Thresholds estimated from training data/domain rules.
6	Encode occupation and other categorical attributes; scale numeric features if required by model.	One-hot encoding for nominal categories; standard/min-max/robust scaling for distance or linear models. Tree models often do not need scaling.	Encoder/scaler fitted on training only.
7	Discretize income only if model/use case benefits; prefer quantile or entropy-based bins over equal-width for skewed data; retain continuous income as an alternative.	Equal-width bins can be sparse/unequal; supervised entropy bins use labels but must be fitted within training folds to avoid leakage.	Yes, fit cut points only on training folds.
8	Feature selection / reduction inside cross-validation; remove redundant variables or use regularization/tree-based selection.	Reduces noise while protecting features that improve minority-class recall.	Yes, training folds only.
9	Train class-aware models: class weights, cost-sensitive learning or training-only resampling such as SMOTE/undersampling.	Addresses imbalance without changing the untouched test distribution. Compare weighted model and resampling.	Resampling only inside training folds.
10	Tune threshold and validate; final evaluation once on untouched test set.	Threshold should reflect business cost and review capacity; report confusion matrix and PR metrics.	Threshold chosen on validation data, not test.

7D – Predict before implementing (1 mark)

Prediction before testing: (1) median/log transformation and robust handling will reduce the influence of the long right tail while preserving legitimate income ranking; (2) stratified training or class-weighted learning will expose the model to more default cases, so the effective training class balance will be less dominated by non-defaults if resampling is used; (3) overall accuracy may decrease slightly, while default recall should increase. Precision may decrease if the model flags more borderline cases. These are hypotheses to verify, not guaranteed outcomes.

7E – Implement and validate (2 marks)

The assignment permits UCI German Credit or a similar public credit dataset. The report should identify the dataset version, target coding, feature columns, missing values and split method. Use the same held-out test rows for both original and redesigned pipelines. The code package accompanying this report contains a Python template that loads a local CSV, constructs a deliberately flawed baseline and a leakage-safe redesigned pipeline, then prints confusion matrices and classification metrics. It does not invent a measured result: run it with the dataset and paste the output and screenshots into the result table.

Metric	Original pipeline – actual run	Redesigned pipeline – actual run	Interpretation
Accuracy	Fill after execution	Fill after execution	Can remain high even with poor recall.
Default precision	Fill after execution	Fill after execution	Fraction of predicted defaulters who truly default.
Default recall	Fill after execution	Fill after execution	Fraction of actual defaulters detected; primary focus.
F1 for default	Fill after execution	Fill after execution	Balance of precision and recall.
PR-AUC	Fill after execution	Fill after execution	Useful under strong class imbalance.
Confusion matrix	Attach output	Attach output	Shows false negatives and false positives.

7F – Defend final design and remaining risk (2 marks)

The redesigned pipeline is preferred if it improves default recall and PR-AUC at an acceptable false-positive workload on the same untouched test data. If recall rises but precision becomes too low, adjust the threshold using expected financial costs and analyst capacity. A remaining risk is population drift: borrower characteristics, interest rates and economic conditions may change after deployment. Monitor feature drift, calibration, default prevalence and subgroup performance, and retrain only under a governed validation process.

CONCLUSION

The assignment demonstrates that mining functionality must match the business question, and that useful patterns depend on reliable KDD preparation. Dispersion measures respond differently to outliers; cleaning must distinguish missing, noisy, inconsistent and invalid values. Integration requires identity and semantic reconciliation, while normalisation is essential for distance-based mining. Data reduction should preserve rare failure patterns, and discretization may be unsupervised or supervised. The bank case shows why leakage-free preprocessing, class-aware learning and minority-class metrics are essential. Manual calculations are shown here; tool validation and screenshots must be generated from the student's own execution.